B.TECH PROJECT SYNOPSIS · BTP–I & BTP–II

# Ledger

A fully local multi-agent data analyst that turns a CSV into insights – and refuses to report the ones that statistics cannot support.

DEPARTMENT

Computer Science & Engineering

INSTITUTION

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DOMAIN

Agentic Systems · Applied Statistics · AutoML

RUNS

Entirely offline · no API key · no data egress

## 01 Abstract

Automated exploratory data analysis is one of the most natural applications of language-model agents: point a system at a spreadsheet and let it tell you what is in there. Every such system built so far shares a defect that is easy to miss and fatal in practice. They test a large number of relationships, report the ones that look interesting, and never account for the fact that searching hard enough through random data will always turn something up.

This is not a subtle flaw. Given a table of pure random noise – where the correct number of findings is exactly zero – a conventional language-model analyst will confidently report a page of insights, each with a plausible narrative attached.

Ledger is a multi-agent analyst built around a single architectural commitment: **the language model proposes, and deterministic statistics decides.** Hypotheses are registered before any test is executed, so nothing can be selected after the fact. Test selection, effect-size estimation and false-discovery-rate correction are performed by code, not by a model. Every sentence in the final report is linked to a ledger entry recording the code that ran, the numbers it returned and the test that justified the claim; a sentence with no entry cannot be written.

The entire system runs offline against a locally-hosted open-weight model. No API key is required and no data leaves the machine, which makes it usable on exactly the confidential datasets – clinical, financial, academic – where automated analysis would be most valuable and is currently least permissible.

## 02 Introduction and motivation

Exploratory data analysis is repetitive, skilled and enormously time-consuming. An analyst loads a table, profiles the columns, forms hunches, tests them, discards most, and writes up the survivors. The mechanical parts of that loop are obvious automation targets, and language models that can write and execute code appear to make it trivial.

The appearance is misleading, for a reason that predates language models entirely. It is the *multiple comparisons problem*. If a relationship is declared significant at `p < 0.05`, then by construction one in twenty tests on data with no real structure will pass. Test a 30-column table exhaustively and there are 435 pairwise relationships available; roughly 22 of them will clear the threshold on pure noise. An agent that reports what it finds will therefore always find something, and will describe it fluently.

Human analysts are taught to guard against this with correction procedures, pre-registration and effect-size reporting. Automated systems have inherited the search behaviour without the discipline, and they operate at a scale that makes the problem far worse: a tireless agent can test thousands of hypotheses in the time a human tests ten.

> THE PREMISE OF THIS PROJECT
>
> Statistical discipline is not a feature to be added to an automated analyst afterwards. It has to be the thing that decides what the analyst is allowed to say. A system that can be talked out of its conclusions by a fluent narrative is not an analyst.

A second motivation is practical. Cloud-hosted analysis tools require uploading the data, which rules them out for precisely the datasets where careful analysis matters most – patient records, financial transactions, student information. Under India's Digital Personal Data Protection Act, transferring such data to a third-party service is a decision with legal weight. A system that runs entirely on the analyst's own machine sidesteps the question.

## 03 Problem statement

> FORMALLY
>
> Given a tabular dataset `D` with no accompanying schema, documentation or analyst guidance, produce a set of findings `F` such that: (a) every `f ∈ F` is supported by a statistical test whose assumptions were checked and whose result was computed by executed code; (b) the family-wise false discovery rate across `F` is controlled at a stated level `q`; (c) every claim in the natural-language report is traceable to a specific computation; and (d) when `D` contains no genuine structure, `F` is empty. All computation occurs locally, with no network access.

Clause (d) is the one that distinguishes this from existing work, and it is the clause most systems fail. It is also the easiest to test: construct a dataset with no real relationships and count what the system claims to have found.

## 04 Literature survey

### Automated analysis and insight discovery

Automated machine learning systems such as Auto-sklearn [1] automate model selection and hyperparameter search, but address prediction rather than explanation. Profiling tools of the ydata-profiling lineage [2] generate exhaustive descriptive statistics with no notion of which findings matter. QuickInsights [3] introduced automatic discovery of salient patterns in multi-dimensional data and explicitly acknowledged the significance-testing problem, applying corrections within its insight types — the closest existing work to this project's concern, though it predates language-model agents and cannot generate or execute novel analysis code.

### Language-model data agents

InsightPilot [4] couples a language model to an insight engine to steer exploration, and Data Interpreter [5] demonstrates end-to-end agentic data-science pipelines with code execution. Both substantially advance what such systems can attempt. Neither controls the false discovery rate across the exploration session, and neither prevents post-hoc selection of hypotheses — the agent is free to run many analyses and narrate the appealing ones.

### Statistical discipline

The Benjamini–Hochberg procedure [6] provides the correction this project applies, controlling the expected proportion of false positives among rejected hypotheses while retaining far more power than family-wise methods. The methodological critique is equally established: *False-Positive Psychology* [7] demonstrated how ordinary analytic flexibility produces significant results from nothing, and the garden-of-forking-paths argument [8] showed that this occurs even without any intent to game the analysis — which is precisely the situation of an autonomous agent. Pre-registration is the standard remedy, and it has not been applied to automated analysts.

### Execution-grounded generation

Self-Debug [9] established that feeding execution results back to a model materially improves generated-code correctness, and evaluation of code models [10] established execution as the only meaningful correctness signal. Ledger extends this principle from code correctness to *claim* correctness: execution decides not merely whether the code ran, but whether the sentence may be written.

## 05 Research gap

- Profiling tools are statistically safe but report everything indiscriminately, leaving the analyst to find what matters.
- AutoML optimises predictive accuracy and says nothing about which relationships are real.
- Language-model data agents can explore and execute freely, but inherit unrestricted multiple testing and post-hoc selection, and therefore cannot distinguish a finding from an artefact of searching.

- No existing automated analyst pre-registers its hypotheses, corrects across the whole session, or can be evaluated on whether it correctly reports *nothing*.
- Systems capable of this analysis are cloud-hosted, which excludes confidential data by construction.

> THE GAP
>
> There is no automated analyst whose architecture makes a false discovery structurally difficult rather than merely unlikely – and none that can be run on data its owner is not permitted to upload.

## 06 Objectives

1. Build a fully offline multi-agent pipeline that ingests an arbitrary CSV and produces a written analysis, with a locally-hosted open-weight model and no network access at any stage.
2. Implement **hypothesis pre-registration**: candidate hypotheses are recorded, with their intended tests, before any test executes; the registry is immutable thereafter.
3. Implement a **deterministic statistical layer** that selects tests from checked assumptions, computes effect sizes with confidence intervals, and applies Benjamini–Hochberg correction across every test run in the session.
4. Implement **claim-level provenance**: a ledger binding every reported sentence to the code executed, the values returned and the test that licensed it, with unbacked sentences rejected before the report is emitted.
5. Build **NULLSET** and **PLANTED**, two synthetic evaluation suites with known ground truth, enabling direct measurement of false discovery rate and statistical power.
6. Evaluate against profiling, single-prompt and execution-enabled agent baselines, and ablate each discipline mechanism to quantify its individual contribution.
7. Deliver a usable local application with an auditable report that a reader can trace from any claim back to the code that produced it.

## 07 System architecture

Six agents in a fixed, hand-written state machine. The orchestration is deliberately not a general-purpose agent framework: every transition must be inspectable, because the ordering constraint between registration and testing is a correctness property, not a convenience. Note which agents involve the language model at all – the ones that decide what is true do not.

| | | |
|---|---|---|
| A1<br>**Profiler**<br>Types, cardinality, missingness, distributions, outliers, candidate keys.<br>DETERMINISTIC | A2<br>**Proposer**<br>Reads the profile, proposes testable hypotheses in natural language.<br>MODEL | A3<br>**Registrar**<br>Freezes hypotheses and their intended tests before execution begins.<br>DETERMINISTIC |
| A4<br>**Executor**<br>Writes pandas code, runs it sandboxed, repairs on failure.<br>MODEL | A5<br>**Statistician**<br>Checks assumptions, picks the test, computes effect size, applies FDR across the session.<br>DETERMINISTIC | A6<br>**Reporter**<br>Writes prose, constrained to surviving claims; unbacked sentences are rejected.<br>MODEL |

■ DETERMINISTIC — DECIDES WHAT IS TRUE

■ MODEL — PROPOSES AND PHRASES, NEVER ADJUDICATES

### The ordering constraint

A3 runs to completion before A4 starts. Once the registry is frozen, no hypothesis may be added, and every registered hypothesis *must* be tested and reported on – including the ones that fail. This is what makes selective reporting impossible: the denominator of the correction is fixed before any result is seen. An agent that could append hypotheses mid-session, or quietly drop the disappointing ones, would invalidate the correction entirely.

### A5 – the statistician

Contains no language model. Given a hypothesis and a data slice it selects the test from checked properties rather than from a model's suggestion:

- Two-group numeric comparison → Shapiro–Wilk and Levene decide between Welch's *t* and Mann–Whitney U.
- Categorical association → expected cell counts decide between $\chi^2$ and Fisher's exact test.
- Numeric association → linearity and outlier structure decide between Pearson and Spearman.
- Every result carries an effect size with a confidence interval – Cohen's *d*, Cramér's *V*, rank-biserial *r* – because significance without magnitude is not a finding.

After all registered tests complete, Benjamini–Hochberg is applied once across the entire family at `q = 0.05`. Survivors become claims. Everything else is recorded as tested-and-not-supported, and appears in an appendix rather than vanishing.

### The ledger entry

```
# one entry per registered hypothesis — the report cannot cite anything else
LedgerEntry {
  hypothesis_id : "H07"
  registered_at : step 3 of 3   // before any execution
  statement     : "tenure differs between churned and retained customers"

  code_executed : "df.groupby('churn')['tenure'].describe()"
  returned      : { churned: n=1869 mean=17.98 sd=19.53,
                    retained: n=5163 mean=37.57 sd=24.11 }

  assumptions   : { normality: FAILED (shapiro p<.001),
                    equal_var: FAILED (levene  p<.001) }
  test_selected : mann_whitney_u   // chosen by A5, not proposed by a model
  statistic     : U = 2_712_444
  p_raw         : 1.2e-197
  p_adjusted    : 5.4e-196        // BH across all 44 registered tests
  effect_size   : rank_biserial r = 0.44  CI95 [0.42, 0.46]

  verdict       : SUPPORTED
  licensed_text : "Churned customers have substantially shorter tenure
                    (median 10 vs 38 months, r = 0.44)."
}
```

The `licensed_text` field is the only text A6 may use for this hypothesis. It cannot strengthen the wording, cannot add a causal reading, and cannot mention a hypothesis that has no entry.

## 08 The null-dataset test

The clearest way to evaluate an automated analyst is to give it a dataset with nothing in it and count what it claims to find. Columns are drawn independently at random, with realistic types, names and missingness so the table is indistinguishable from a real one by inspection. The correct number of findings is zero, and it is zero by construction rather than by judgement.


**Figure 1 – Expected result, not measured.** These are the values the experiment in §10 is designed to test, stated in advance so the project can be judged against a prediction rather than a post-hoc story. Bars are outlined rather than filled to mark them as hypothesised. If the full system does not land at or near zero, that is a reportable negative result about the architecture.

The same protocol run in reverse gives the power measurement. **PLANTED** datasets contain a known number of genuine relationships at controlled effect sizes, buried among noise columns. Sweeping effect size and sample size traces the boundary where the system stops detecting real structure – because a system that reports nothing is trivially safe and useless, and the interesting question is what discipline costs in sensitivity.

## 09 Novelty and contributions

1. **Session-level false discovery control in an autonomous analyst.** Correction is applied once across every hypothesis registered in the session, not per-test and not per-insight-type.
2. **Hypothesis pre-registration for agents.** Importing a methodological safeguard from empirical science into agent architecture, enforced by an ordering constraint the orchestrator cannot violate.
3. **A deterministic adjudication layer.** The component that decides what is true contains no language model; test selection follows from checked assumptions.
4. **Claim-level provenance with licensed text.** The reporter can only emit sentences the statistician has licensed, making unbacked narration structurally impossible rather than merely discouraged.

5. **NULLSET and PLANTED**, an evaluation protocol measuring what automated analysts have never been measured on: whether they correctly report nothing, and what discipline costs in power.
6. **A fully offline analyst** usable on data that may not legally be uploaded.

## 10 Evaluation plan

DATASETS

| SUITE | SCALE | CONSTRUCTION | MEASURES |
|---|---|---|---|
| **NULLSET** | `200 tables` | Independently drawn columns, realistic names, types and missingness; zero real relationships | **False discovery rate** – the headline number |
| **PLANTED** | `300 tables` | Known relationships at controlled effect sizes among noise columns; sample size swept | Statistical power; effect-size estimation error |
| **REAL-KNOWN** | `~20 datasets` | Public datasets with well-documented established findings | Rediscovery rate of known facts |
| **REAL-WILD** | `~30 datasets` | Unseen public datasets; two independent analysts adjudicate the reports | Claim groundedness; expert agreement |

BASELINES

| BASELINE | WHAT IT ISOLATES |
|---|---|
| ydata-profiling | Statistically safe but undirected – the no-LLM floor |
| Single-prompt LLM on the raw CSV | No execution at all; measures pure narration |
| LLM + code execution, no discipline | **The critical comparison** – this is what current agent frameworks do |
| Ledger without FDR correction | The contribution of correction alone |
| Ledger without pre-registration | The contribution of the ordering constraint alone |
| Ledger with an LLM statistician | Whether determinism in A5 actually matters |

Two further metrics matter and are cheap to collect. **Groundedness**: the proportion of sentences in the final report traceable to a ledger entry, which should be exactly 1.0 by construction and is therefore a correctness check on the implementation rather than a research result. **Reproducibility**: run the system twice on the same input and measure whether the conclusions agree – a property no existing agent reports, and one that is likely to embarrass the baselines.

## 11 Tools and technologies

| LAYER | CHOICE | RATIONALE |
|---|---|---|
| Model host | Ollama, local | No API key, no network, no data egress – a requirement, not a preference |
| Model | Qwen2.5-Coder-7B-Instruct (4-bit); 1.5B fallback for CPU-only machines | Code generation is the only task asked of it; 7B in 4-bit fits a T4 and a decent laptop GPU |
| Orchestration | Hand-written state machine | The registration-before-execution ordering is a correctness property; a general agent framework cannot guarantee it |
| Execution sandbox | Subprocess isolation, `rlimit`, import allow-list, no sockets | Model-generated code runs against the user's real data |
| Data | pandas, NumPy, PyArrow | Standard; PyArrow for larger-than-memory CSVs |
| Statistics | SciPy, statsmodels, pingouin | `statsmodels.stats.multitest` supplies Benjamini–Hochberg; pingouin supplies effect sizes with intervals |
| Interface | Streamlit | Local-first, single command to launch, no separate frontend to maintain |
| Report | HTML with expandable ledger entries; PDF export | Every claim clicks through to the code that produced it |

No component requires training, fine-tuning or an internet connection. The heaviest operation in the system is a forward pass through a 7B model, and the entire pipeline runs on a laptop.

## 12 Expected outcomes

- A working offline application: drop in a CSV, receive an audited analysis with every claim traceable to executed code.
- A quantified false discovery rate for current agentic analysis approaches – a number the field does not currently have.

- **NULLSET** and **PLANTED** released publicly as a reusable evaluation protocol for automated analysts.
- A power analysis establishing what statistical discipline costs in sensitivity, so the trade-off is stated rather than assumed.
- An ablation attributing the result to specific mechanisms – correction, pre-registration, deterministic adjudication – rather than to the system as a whole.
- A paper for a data-management, applied-ML or software-engineering venue.

## 13 Project timeline

BTP–I – PIPELINE, SANDBOX, EVALUATION HARNESS

| WEEKS | MILESTONE | DELIVERABLE |
|---|---|---|
| 1–2 | Literature survey; local model benchmarking on pandas code generation | Survey; model selection with evidence |
| 3–4 | A1 profiler; execution sandbox with resource limits | Profiler output on 20 real datasets |
| 5–6 | A2 proposer and A4 executor with repair loop | End-to-end unaudited pipeline |
| 7–9 | NULLSET and PLANTED generators; baseline harness | **Baseline FDR numbers** – the motivating result |
| 10–12 | A3 registrar; ordering constraint enforced and tested | Pre-registration working; first ablation |
| 13–14 | BTP-I report and mid-term demonstration | Report; live CSV-to-analysis demo |

BTP–II – DISCIPLINE LAYER, EVALUATION, PUBLICATION

| WEEKS | MILESTONE | DELIVERABLE |
|---|---|---|
| 1–3 | A5 statistician: assumption checks, test selection, effect sizes | Deterministic adjudication layer |
| 4–5 | Session-level Benjamini–Hochberg; ledger and licensed text | **The null-dataset result** |
| 6–7 | A6 reporter with groundedness enforcement | Auditable report; groundedness = 1.0 verified |
| 8–10 | Full evaluation: power curves, real datasets, expert adjudication | Complete results and ablations |
| 11–12 | Streamlit application; PDF export; packaging | Public release build |
| 13–16 | Paper writing; benchmark release; final viva | Manuscript, released suites, thesis |

## 14 Risk analysis

| RISK | IMPACT | MITIGATION |
|---|---|---|
| Local 7B model writes incorrect pandas code | `High` | Repair loop against real execution errors; a library of parameterised templates for the twenty most common analyses, with free-form generation only as fallback. Template coverage is itself a reported metric. |
| Discipline makes the system too conservative to find anything | `High` | PLANTED power curves quantify this directly; `q` is a user-facing dial, and the cost of discipline is a headline result rather than a hidden failure |
| Hypothesis registry explodes combinatorially on wide tables | `Medium` | Registration budget fixed by column count; the proposer must prioritise before freezing, and the budget is recorded in the report |
| Model-generated code escapes the sandbox | `Medium` | Import allow-list, no socket access, resource limits, read-only data mount; the sandbox is tested adversarially |
| Local inference too slow for interactive use | `Medium` | Batch the proposer, cache by content hash, 1.5B fallback; wall-clock is reported honestly rather than engineered around |
| Scope creep into causal inference | `High` | Explicitly out of scope and stated as such in the report the system emits; every claim is worded associationally by construction |

## 15 Threats to validity

- **Synthetic nulls may be easier than real ones.** Real data contains genuine but uninteresting structure – encoding artefacts, collection effects – that NULLSET lacks. Mitigated by including permuted real datasets, which preserve marginal distributions while destroying relationships.
- **FDR control assumes the registry is the full family.** If the proposer's choice of hypotheses is itself informed by peeking at the data, the correction is optimistic. The profiler's output is deliberately restricted to marginal summaries for this reason, and the restriction is enforced in code.
- **Expert adjudication on REAL-WILD is subjective.** Two independent annotators with reported agreement, and disagreements published rather than resolved silently.
- **Results are model-dependent.** Every experiment is repeated across at least two local models so conclusions are not artefacts of one set of weights.

# References


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